Linear Regression Analysis: Aligning Host Pricing Decisions with Consumer Preferences in Amsterdam, North Holland, Netherlands

Introduction

Airbnb's low cost and variety attract users, and price often signals quality (Wang & Jeong, 2018). However, it remains unclear whether the listing prices set by Airbnb hosts (Response variable) truly reflect the factors (predictors) that consumers may consider important when choosing accommodations, such as room type, guest reviews, and host responsiveness. Therefore, this proposal examines whether Airbnb hosts' pricing decisions align with consumer preferences and whether Airbnb listings are reasonably priced for consumers. Specifically, addressing the research question of “to what extent do Airbnb hosts’ pricing decisions reflect consumer preferences, as represented by host attributes, customer review metrics, and different property features, in Amsterdam, North Holland, Netherlands?”

A multiple linear regression is used to answer this question since it quantifies the linear relationship between price and consumer preferences while maintaining high interpretability. By estimating the direction and magnitude of coefficients, consumers identify whether host pricing decisions are consistent with consumer preferences. *P-values* also indicate whether these preferences have a significant impact on price. Furthermore, confidence intervals are used to assess the reliability of each factor’s effect. If this interval includes zero, it indicates inconsistency with consumer preferences. Moreover, predictions can help identify price differences across different room types while controlling for host responsiveness and customer reviews.

Three research papers have examined the determinants of Airbnb listing prices, emphasizing the roles of host–guest relationships, review metrics, and room type. Chang and Li (2021) find that hosts’ friendliness and responsiveness enhance guest satisfaction and willingness to pay. Teubner et al. (2017) show that higher review scores and more reviews act as trust signals, leading to higher prices. Together with Wang and Jeong (2018), they also found that entire homes are typically priced higher than private rooms, which are priced higher than shared rooms. Wang and Jeong (2018) further reveal a positive relationship between a property’s accommodation capacity and its price. These findings justify the chosen predictors and support applying multiple linear regression to explore their effects.

Both hosts and guests of Airbnb gain from this model. Customers can judge if prices meet expectations and spot overpriced listings by using the model’s prediction. By identifying the characteristics that have the most significant impact on price, the model helps hosts modify their pricing strategies. While hosts in Amsterdam significantly consider the private room, and *accommodates* when listing price, they seem to pay little attention to customer reviews, entire home, and shared room, host–guest relationships suggesting that their pricing decisions only partly match what consumers value.

Data description

Dr. Philip E. Cannata, an Oracle Principle Software Engineer, uploaded the Airbnb dataset to *data.world* (University of Texas at Austin, n.d.). The dataset, sourced from *Inside Airbnb* (n.d.), was compiled from publicly available Airbnb listings and curated for educational use (Cannata, 2017). By combining related variables for analysis, the cleaned dataset, which focuses on listings in North Holland, Netherlands, keeps only pertinent columns.

Table 1: Statistical Summary of the Listing Prices (Response) in Amsterdam, North Holland, Netherlands

Statistic	Value
Min.	19.0000
1st Qu.	82.0000
Median	105.0000
Mean	124.7278
3rd Qu.	149.0000
Max.	1400.0000

Listing price is a suitable dependent variable for linear regression as it is ordered, continuous and quantitative. As shown in *Table 1*, prices are concentrated around the interquartile range (82–124.7) with a median of 105, indicating an approximately normal distribution. Although a host may manage multiple listings, each property has a unique ID, ensuring that observations are independent (Cannata, 2017).

Table 2: Summary Statistics for Consumer Preferences (Predictors)

Variable	Mean	SD	Median	Range	Skew
Accommodates	3.133	1.754	2	15.00	3.102
Private_Room	0.184	0.387	0	1.00	1.632
Shared_Room	0.005	0.072	0	1.00	13.783
Accommodates:Private_Room	0.419	1.051	0	16.00	5.104
Accommodates:Shared_Room	0.016	0.345	0	16.00	37.943
Number_of_Reviews	18.463	28.225	8	296.00	3.616
Review_Scores_Rating	93.370	7.371	95	80.00	-2.757
# Reviews:Review_Scores_Rating	1724.034	2643.273	784	26413.00	3.642
Host_Response_Rate	0.910	0.148	1	0.98	-2.164
hr_within_hour	0.300	0.458	0	1.00	0.871
hr_within_few_hours	0.384	0.486	0	1.00	0.476
hr_few_days_or_more	0.024	0.152	0	1.00	6.279

For this proposal, data cleaning yielded 5,599 valid listings in Amsterdam, North Holland, Netherlands, after removing missing and invalid values. In *Table 2*, skew shows that most predictors have a few large outliers due to right-skewed tails, particularly in shared room and response times measured in days. This suggests that most listings are not shared rooms, as they generate lower profits, consistent with Wang and Jeong (2018). Most hosts respond promptly. The large variance and range in customer review metrics suggest substantial differences in guest satisfaction across listings, with some listings attracting extremely high or low reviews.

Figure 1. Interaction between Accommodates and Private Room

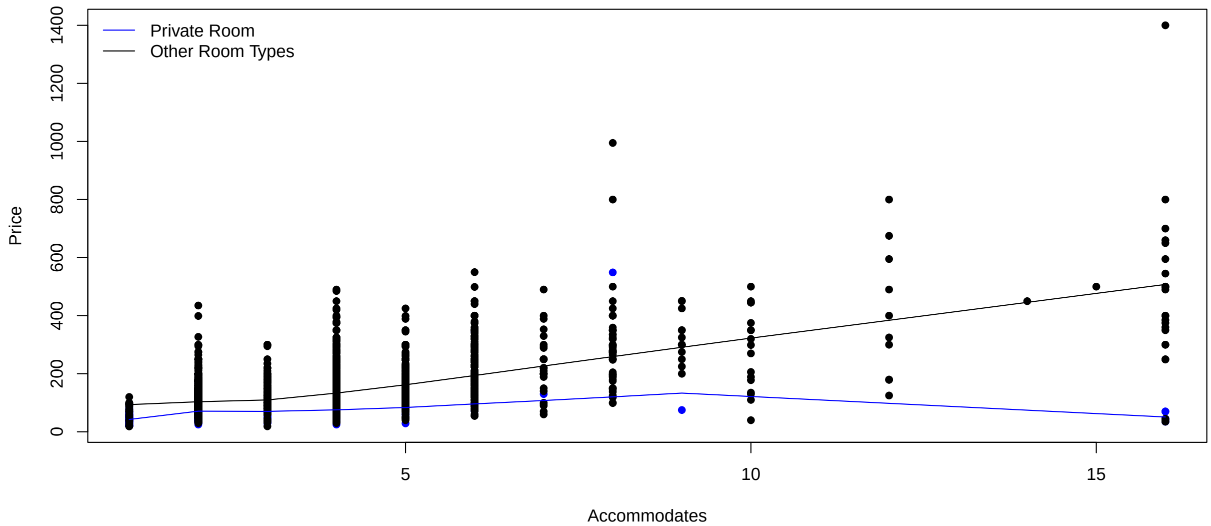
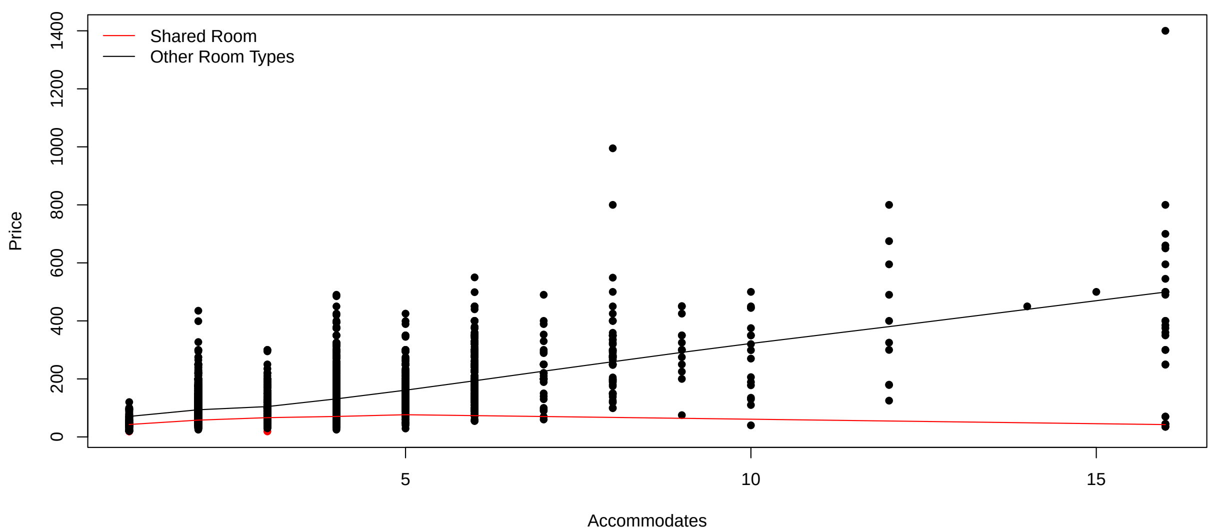


Figure 2. Interaction between Accommodates and Shared Room



Based on both *Figure 1* and *Figure 2*, the relationship between the number of guests a listing can accommodate and its price differs by room type. The slopes for private room and shared room are flatter and even slightly negative compared to other room types, indicating the effect of *accommodates* capacity on price depends on the room type. The interaction between *number_of_reviews* and *review_scores_rating* is supported by Teubner et al. (2017), captures how higher ratings become more credible when backed by many reviews, jointly influencing price.

Ethics Discussion

Our dataset is taken from publicly available Airbnb listings in Amsterdam, North Holland. I believe it is trustworthy because it follows the basic ideas of ethical data use such as accuracy, transparency, and fairness. The data were collected from open sources, not private platforms, and they contain only non-personal information. Each record describes property and host features instead of private details, which keeps people's privacy safe and makes the dataset suitable for academic analysis.

During the data cleaning process, we removed missing values and converted all prices and review scores into clear numeric forms. This helped make the dataset more consistent and reduced possible errors. We also kept all values close to their original form, without making changes that could distort the meaning. These steps reflect honesty and integrity in research, which were emphasized in the ethics module.

Our use of this dataset is ethical because it is for learning and research, not for business or profit. We are exploring how host and property features influence Airbnb prices, to understand consumer behavior in the sharing economy. All results are shown at a general level, and no personal names or identifiers are included. This approach respects privacy, consent, and data ownership, ensuring that our study does not harm any individuals.

In conclusion, our dataset and methods meet the principles of responsible and ethical research. The data are open, anonymous, and handled carefully. By keeping transparency, fairness, and respect in our analysis, our project supports academic honesty and contributes to a better understanding of how data can be used responsibly in research.

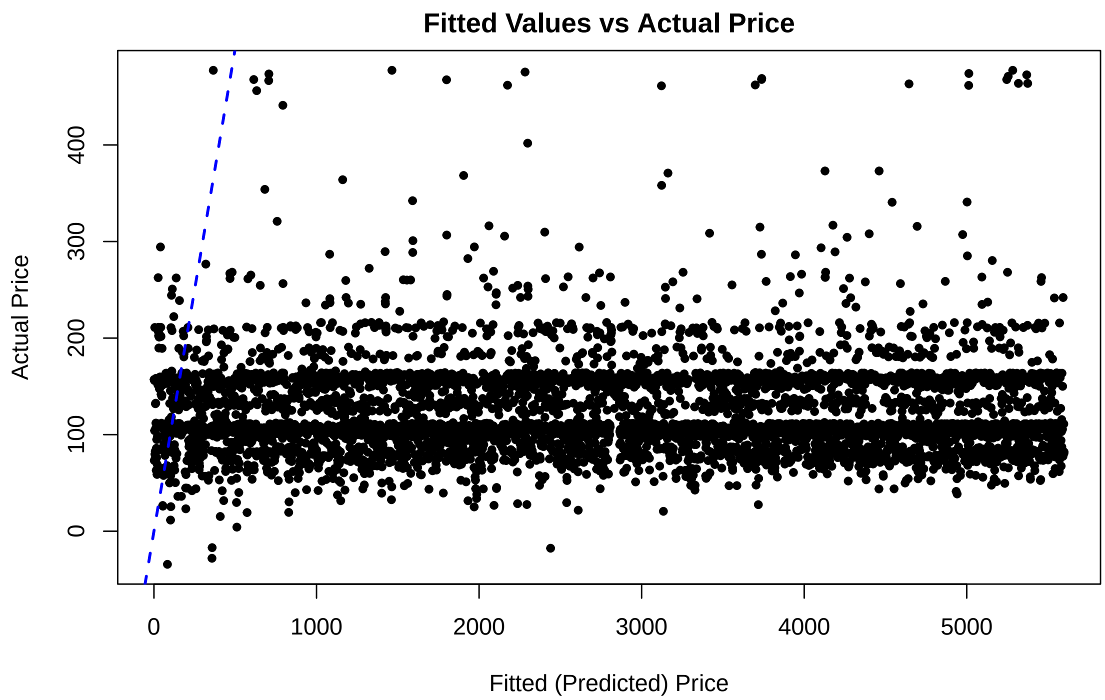
Preliminary results

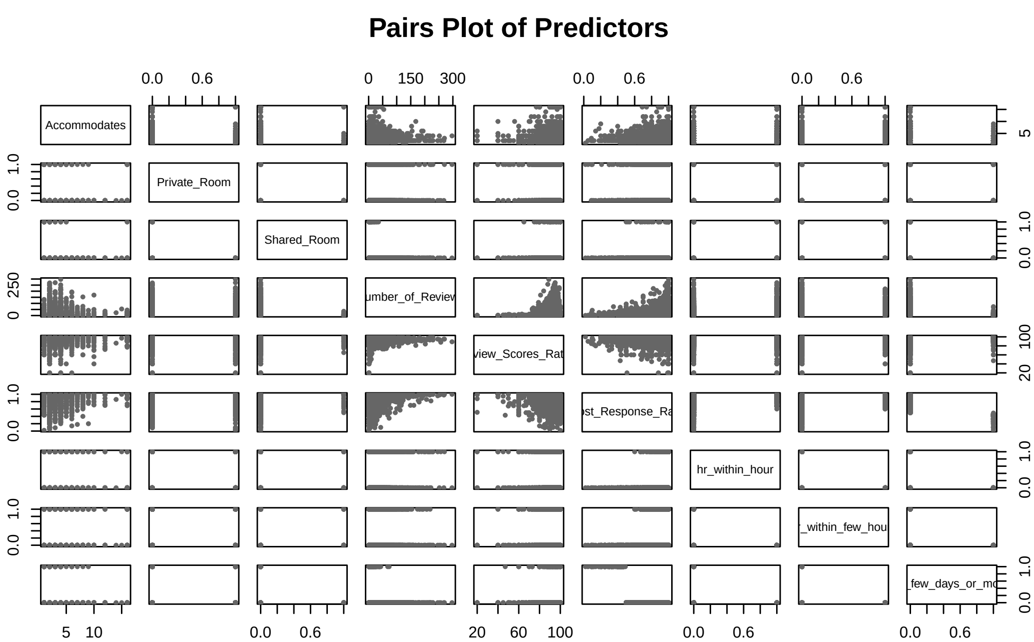
The preliminary model explores whether Airbnb hosts price their listings according to consumer preferences. *Accommodates* has the most significant effect, indicating that customers have a strong willingness to pay for capacity. The estimated coefficients further reveal that hosts think private room commands a higher premium than others because of its privacy. The negative interactions between *accommodates* and *room type* imply that prices for private and shared rooms increase less with each additional guest than entire homes, aligning with Wang and Jeong's (2018) conclusion of the entire home with the highest listing price.

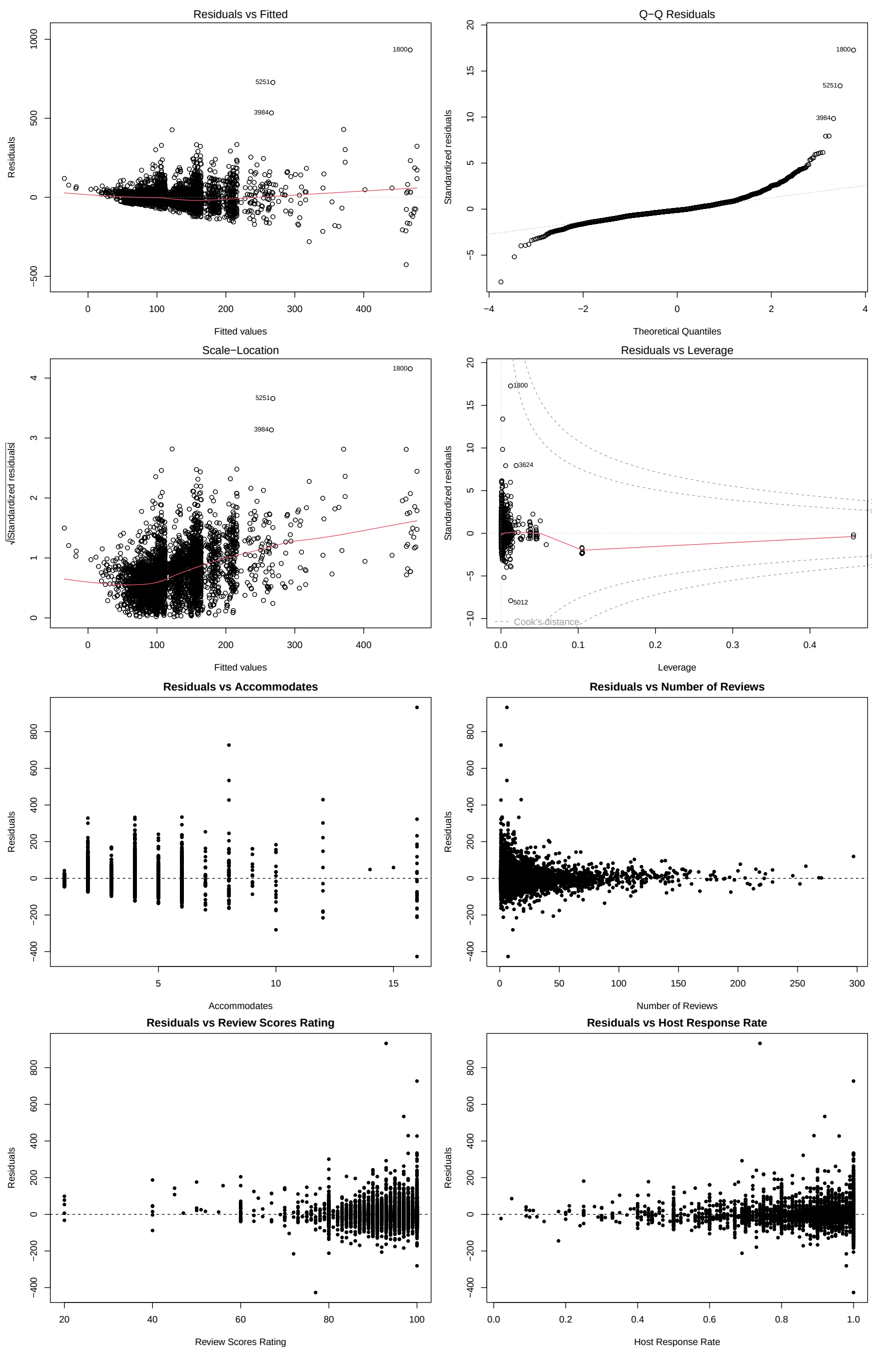
Table 3: Estimated Coefficients and 95% Confidence Intervals

Variable	Estimate	Std. Error	t value	p value	CI lower	CI upper
(Intercept)	11.159	12.019	0.928	0.353	-12.402	34.721
Accommodates	26.144	0.460	56.780	0.000	25.242	27.047
Private_Room	16.643	3.818	4.359	0.000	9.158	24.127
Shared_Room	9.052	13.175	0.687	0.492	-16.776	34.879
`Accommodates:Private_Room`	-20.326	1.355	-15.003	0.000	-22.982	-17.670
`Accommodates:Shared_Room`	-26.214	2.760	-9.500	0.000	-31.624	-20.805
Number_of_Reviews	-4.257	0.561	-7.585	0.000	-5.357	-3.157
Review_Scores_Rating	0.392	0.111	3.534	0.000	0.174	0.609
`#Reviews × Rating`	0.043	0.006	7.198	0.000	0.031	0.055
Host_Response_Rate	4.182	7.383	0.566	0.571	-10.291	18.655
hr_within_hour	-0.403	2.266	-0.178	0.859	-4.845	4.038
hr_within_few_hours	4.445	2.036	2.183	0.029	0.453	8.437
hr_few_days_or_more	-9.925	6.024	-1.648	0.100	-21.734	1.885

Although the intercept (entire home) and shared room coefficients have confidence intervals including zero, other predictors exhibit distinct and significant relationships with price, as shown in Table 3. Furthermore, the estimated coefficient of *number-of-reviews* is negative, possibly reflecting that hosts lower prices to attract more reviews. The coefficients for *review-scores-rating* and its interaction with *number-of-reviews* are small, suggesting that host attentiveness and review quality have a weaker influence on price than expected. This result contrasts with Teubner et al. (2017), who reported a stronger price premium associated with highly rated and responsive hosts. Regarding host-related variables, *host response rate* shows a positive but insignificant relationship with price, suggesting that higher response rates alone do not ensure a price premium. The *host-response-time* dummies show that listings responding within a few hours are priced slightly higher than the baseline "within a day," while responses within an hour or after a few days show weaker or negative effects, indicating moderate promptness may be most appealing to guests.







The residual-versus-fitted and pair plots show approximately linear relationships, with no serious multicollinearity. Some predictors display mild heteroskedasticity, with the Scale–Location plot showing increasing variance for high-priced listings. The QQ plot suggests near-normal residuals with a right-tail deviation from luxury outliers. Overall, the assumptions remain reasonably satisfied, with minor variance and normality concerns that may slightly reduce prediction accuracy.

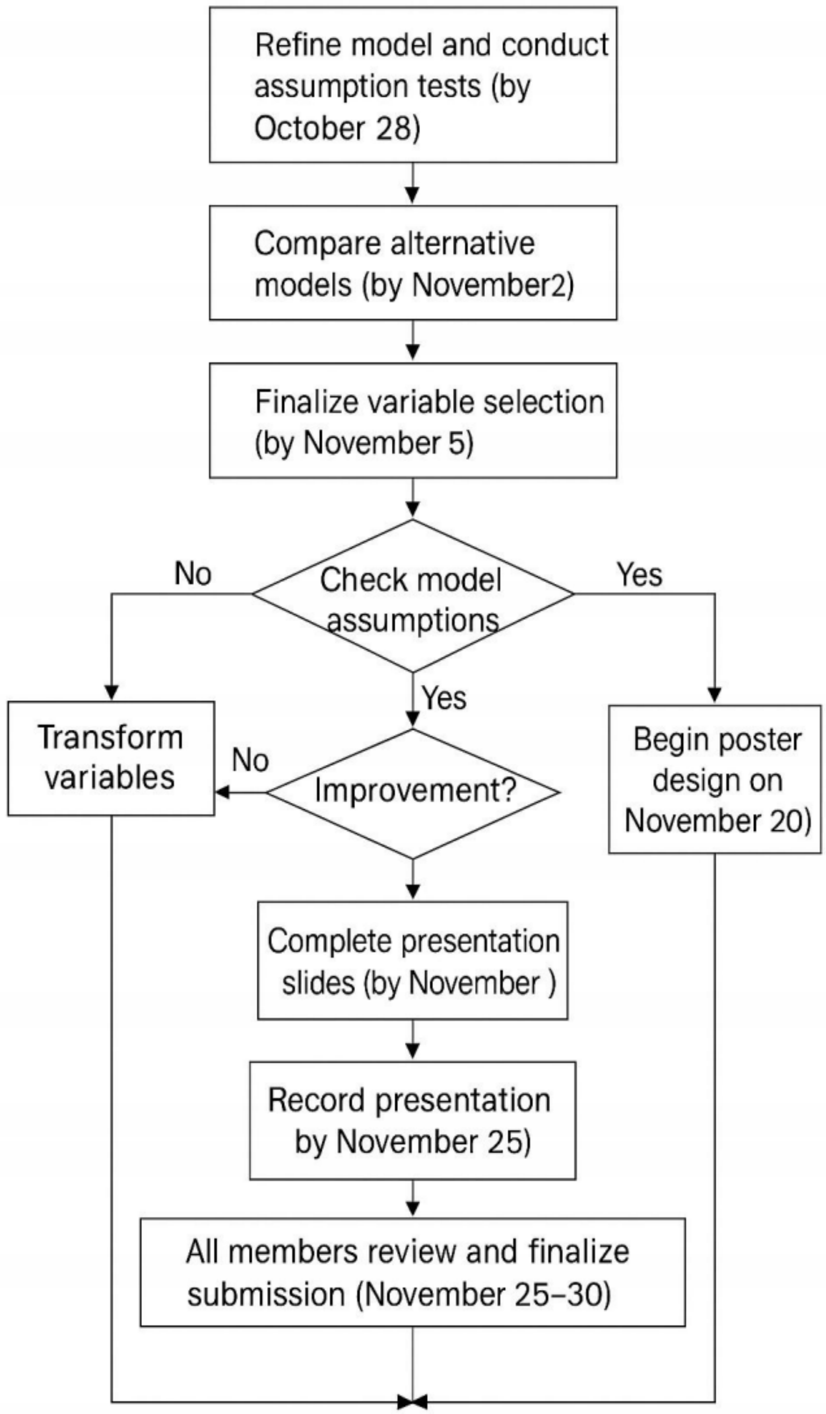
Plan

Our group aims to improve the current regression model by examining whether Airbnb hosts set listing prices according to consumers' preferences. We will begin by reviewing all predictors to identify any that are redundant or irrelevant. If two predictors convey similar information, we will retain the more meaningful one or combine them. Next, we will conduct hypothesis tests to identify significant predictors and verify whether their coefficients align with those of previous studies. Core variables such as *accommodates* and *room-type* will remain in the model because they capture how customers value space and privacy when choosing an Airbnb listing.

Throughout the process, we will regularly check model assumptions by examining the linear relationships between the response and predictors and testing residuals for normality and constant variance. If residuals show heteroscedasticity or right-skewness, we will apply a log transformation to the dependent variable (price) to stabilize variance and improve model fit. We may also transform skewed predictors, such as *number_of_reviews*, to make relationships more linear and interpretable. Additional checks will ensure that no variable dominates the model and that multicollinearity stays within acceptable limits.

Model performance will be evaluated using adjusted R^2, *p-values*, and residual diagnostics. Predictors will be added or removed step by step, comparing changes in explanatory power and interpretability. We will balance simplicity and accuracy to avoid overfitting, guided by insights from previous research to keep the model statistically valid and contextually meaningful. Starting from October 28, our group will refine the model and test assumptions. By November 2, we will finalize variable selection, then design the poster from November 10 and complete the slides by November 20. The presentation will be recorded by November 25, and the project reviewed before final submission on November 30. Lu Chen will write and edit, Karen will handle R coding and visualization, and Jiajia Lu will interpret results and link them to literature.

Flow Chart


Refine model and conduct assumption tests (by October 28)
Compare alternative models (by November2)
Finalize variable selection (by November 5)
Check model assumptions
No
Yes
Yes
Transform variables
No
Improvement?
Begin poster design on November 20)
Complete presentation slides (by November)
Record presentation by November 25)
All members review and finalize submission (November 25–30)

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